Wire Colors:

Manufactured Wire Harness: Colors are

Power: Red

Ground: GRAY

ANY OTHER wire is white

4 DT Pin Connector: Manmade Harness Colors

- 12V Pin: Red

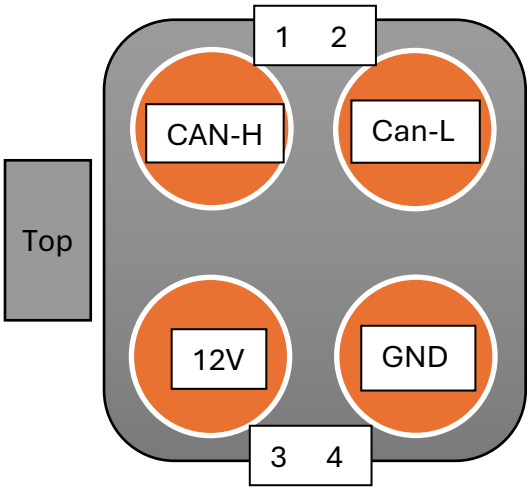
- GND Pin: Black

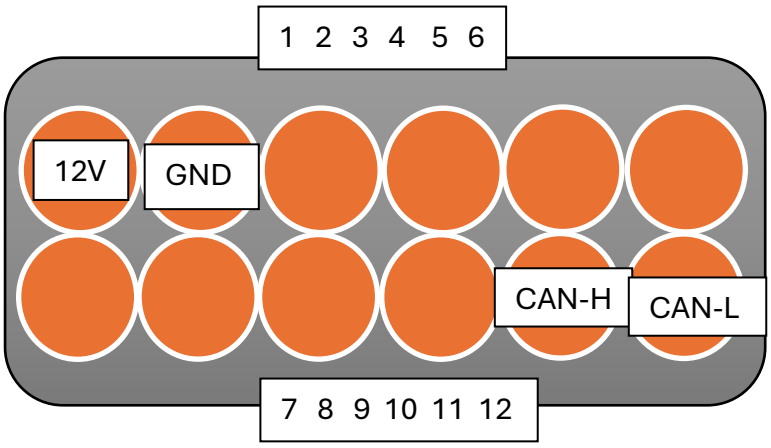
- CAN-H: Green

- CAN-L: Blue

12 DT Pin Connector: (see diagram)

Connector and Pinout:





CAN Bit Rate:

-1 Mbit/s

CAN ID blocks:

-IDs will be claimed here, which are pulled from the registry

-Boards **cannot** share an ID, each one is unique to each board

CAN ID Blocks:

-The order will be based off how many bits each ID is and will be able to be identified within our system.

-Priority, Data, etc...

Header and Heartbeat Format:

-**Frame Header**:

Byte(s) 0: Sequence counter

Byte(s) 1-2: µs since Sync

Byte(s) 3-7: The data

-**Heartbeat**:

Syncs on 0x080 every 50 ms Heartbeat on 0x700 + the node ID once every second, if not read for 3 seconds will eventually trigger an alert that will also eventually appear on the dashboard.

Termination Rule:

-120Ω Resistor: on each end of buggy **within** the wire harness.

DAQ at on end REAR STEER AT OTHER

12 Volt Power:

-Goldenmate 12V LiFePO4 Battery.

MCU:

-Esp32-S3

-3.3V CAN Transceiver